

# Air Conditioners Data Analysis

## Stage 8: Results & Interpretation

This notebook is independently executable in Google Colab. Every step explains what we do, why we do it, what we obtain, and how the result is used in the next stage.

### Stage Objective

Stage 8 brings together the dataset profile, analysis results and visual evidence into final project findings.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# Load the project CSV directly so this notebook is independently executable.
csv_path = "/content/Air_Conditioners_Cleaned.csv"
df = pd.read_csv(csv_path)

print("CSV loaded successfully.")
print("Dataset shape:", df.shape)
```

CSV loaded successfully.  
Dataset shape: (720, 14)

### Why This Stage Matters

A project is not complete when charts are generated. The final stage explains what the charts and statistics mean and how they connect to the original objective.

```
print('Missing values:',int(df.isna().sum().sum()))
```

Missing values: 217

### Reload the CSV

The final notebook loads the same supplied CSV directly so the reported dataset size and quality can be verified.

```
print('Duplicate rows:',int(df.duplicated().sum()))
```

Duplicate rows: 0

### Final Dataset Profile

The project is based on **720 rows and 14 columns** in the supplied CSV. This profile anchors the conclusions to the actual dataset.

```
nums=df.select_dtypes(include=np.number).columns.tolist(); print(df[nums].describe().T)
```

|                  | count | mean         | std          | min           | \ |
|------------------|-------|--------------|--------------|---------------|---|
| ratings          | 720.0 | 3.888194     | 0.612151     | 1.000000      |   |
| no_of_ratings    | 720.0 | 151.262500   | 562.513005   | 1.000000      |   |
| discount_price   | 720.0 | 40183.979167 | 11938.526137 | 199.000000    |   |
| actual_price     | 720.0 | 55479.741139 | 18984.776512 | 499.000000    |   |
| discount_amount  | 720.0 | 15295.761972 | 13102.876643 | -38541.000000 |   |
| discount_percent | 720.0 | -2.351720    | 262.575704   | -5938.520801  |   |
| tonnage          | 626.0 | 1.468164     | 0.501664     | 0.750000      |   |
| energy_star      | 597.0 | 3.577889     | 0.951534     | 1.000000      |   |

|                  | 25%          | 50%          | 75%          | max           |
|------------------|--------------|--------------|--------------|---------------|
| ratings          | 3.900000     | 4.000000     | 4.000000     | 5.000000      |
| no_of_ratings    | 8.000000     | 18.000000    | 40.000000    | 9577.000000   |
| discount_price   | 37282.500000 | 39190.000000 | 41999.250000 | 128800.000000 |
| actual_price     | 51472.500000 | 56990.000000 | 61990.000000 | 149000.000000 |
| discount_amount  | 12726.750000 | 17800.000000 | 20403.250000 | 65000.000000  |
| discount_percent | 25.013559    | 31.23355     | 34.526545    | 60.12024      |
| tonnage          | 1.000000     | 1.500000     | 1.500000     | 8.500000      |
| energy_star      | 3.000000     | 3.000000     | 5.000000     | 5.000000      |

### Final Quality Review

Missing-value and duplicate checks are repeated so the final discussion is based on a verified data state.

```
cats=df.select_dtypes(include=['object','category','bool']).columns.tolist()
for c in cats[:5]: print(c,df[c].value_counts().head(10).to_dict())
```

```
name {'Samsung 1.5 Ton 3 Star Wind-Free Technology Inverter Split AC (Copper, Convertible 5-in-1 Cooling Mode, Anti-bacterial Fil...': 3, 'White Westinghouse 1.5
main_category {'appliances': 720}
sub_category {'Air Conditioners': 720}
image {'https://m.media-amazon.com/images/I/51JFb7FctDL.AC UL320 .jpg': 6, 'https://m.media-amazon.com/images/I/417b-AztnjL.AC UL320 .jpg': 5, 'https://m.media
link {'https://www.amazon.in/Portable-Conditioner-Personal-Evaporative-Display/dp/B09YRMFZGJ/ref=sr_1_70?qid=1679134274&s=kitchen&sr=1-707': 1, 'https://www.ame
```

### Key Numerical Findings

The main descriptive statistics and group summaries are reviewed to identify the strongest observable patterns.

```
print(df['discount_price'].describe().to_dict() if 'discount_price' in df.columns else 'Price field not detected')
```

```
{'count': 720.0, 'mean': 40183.979166666664, 'std': 11938.526137340927, 'min': 199.0, '25%': 37282.5, '50%': 39190.0, '75%': 41999.25, 'max': 128800.0}
```

## ✓ Key Category Findings

Category counts and group-level measures are interpreted together rather than looking at one statistic in isolation.

```
print(df['ratings'].describe().to_dict() if 'ratings' in df.columns else 'Rating field not detected')
```

```
{'count': 720.0, 'mean': 3.888194444444443, 'std': 0.6121505591456029, 'min': 1.0, '25%': 3.9, '50%': 4.0, '75%': 4.0, 'max': 5.0}
```

## ✓ Visual Evidence

The charts from Stage 7 are connected to the corresponding numerical results, showing how each interpretation was obtained.

```
print('Final interpretation connects numerical summaries with the visual evidence from Stage 7.')
```

```
Final interpretation connects numerical summaries with the visual evidence from Stage 7.
```

## ✓ Practical Meaning

The analysis provides a structured understanding of the air-conditioner products represented in the dataset, including observable differences in available specifications and commercial attributes.

```
print('Limitations: conclusions depend on the supplied dataset and available attributes.')
```

```
Limitations: conclusions depend on the supplied dataset and available attributes.
```

## ✓ Limitations and Future Scope

The conclusions depend on the supplied dataset, its available attributes and its represented products. Future work could add more data, interactive dashboards, richer features and predictive analysis.

```
print('Future scope: larger data, dashboards, richer features and predictive modelling.')
```

```
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```